Jaymin Sheladia

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Education

University of Southern California — Los Angeles, USA
Master of Science in Computer Science — January 2026 – December 2027

Graphic Era Deemed to be University — Dehradun, India
B.Tech in Computer Science and Engineering — October 2021 – June 2025

Technical Skills

Languages: C, C++, Python, Java, JavaScript, HTML, SQL
Frameworks/Libraries: Angular, React, Express.js, Spring Boot, Flask, TensorFlow, Keras, OpenCV, Scikit-learn
Developer Tools: Git, GitHub Actions, Docker, VS Code, PyCharm, Jupyter Notebook, Postman
Databases: MySQL, MongoDB, PostgreSQL
Concepts: Data Structures & Algorithms, Distributed Systems, Unix/Linux, RESTful API Design, CI/CD, Reinforcement Learning, Time Series Analysis
Relevant Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Applied Natural Language Processing, Agentic AI Systems
Certifications: IBM Machine Learning Professional Certificate (Coursera, 2024)

Work Experience

Cognizant Technology Solutions — May 2025 – September 2025
Programmer Analyst Trainee – Software Engineering — Coimbatore, India

- Engineered a full-stack Travel Planner and Booking System, designing **10+** RESTful APIs in Spring Boot and Angular to modernize manual booking workflows and centralize itinerary, booking, and preference management.
- Integrated a Generative AI-based recommendation engine that analyzes **3** key factors – user interests, budget, and trip duration – to surface personalized suggestions that simplified the booking decision process.

CODTECH IT SOLUTIONS — June 2024 – August 2024
Machine Learning Intern — Remote

- Designed and trained a fraud detection model to flag fraudulent transactions, applying anomaly detection and supervised learning on over **200,000** credit card transactions to reach **98%** accuracy.
- Optimized algorithms and rebuilt data preprocessing pipelines for existing production models, enhancing their accuracy by **12–15%** across multiple live machine learning projects and reporting dashboards.

Projects

Model Router | GitHub | Python, FastAPI, SQLite — July 2026 – August 2026

- Built a policy-aware LLM router that extends prompt-complexity routing, as used by tools like RouteLLM, with identity, per-project authorization, budget enforcement, and a full audit trail explaining every decision.
- Measured the routing classifier against a hand-labeled oracle set using live model calls, reaching **100%** classification accuracy and cutting cost by **27.9%** versus always routing to the top-tier model.

AI Research Assistant | GitHub | Python, FastAPI, Next.js, RAG — June 2026 – July 2026

- Architected a full-stack research assistant that lets researchers chat with papers, surface related work, and navigate a knowledge graph instead of manually cross-referencing citations.
- Benchmarked the ingestion pipeline against live large language model calls, achieving end-to-end processing in under **4.3** seconds per paper, with LLM inference accounting for **85–95%** of total latency.

Sequence Alignment: DP vs. Divide & Conquer | GitHub | Python, Algorithms — March 2026 – April 2026

- Tackled the $O(m \times n)$ memory bottleneck of classical Needleman-Wunsch dynamic programming by implementing Hirschberg's memory-efficient divide-and-conquer algorithm for global DNA sequence alignment.
- Reduced memory usage from $\mathbf{O(m \times n)}$ to $\mathbf{O(m+n)}$, cutting peak memory consumption by up to **90%** on large inputs while preserving identical optimal alignments.

Research

Deep Learning Approach for Malicious URL Detection | Co-author; Published Research Paper — 2025

- Compared **4** deep learning architectures (CNN, RNN, LSTM, Bi-LSTM) for malicious URL detection, engineering character-level preprocessing pipelines on the IEEE Dataport dataset for sequential analysis at scale.
- Achieved **99.25%** accuracy with the Bi-LSTM model, outperforming CNN (**86.5%**), LSTM (**63.75%**), and RNN (**53%**) baselines across standard classification and error-rate metrics.